QP Code **Reg. No……**

Name……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION OCTOBER 2026
Fifth SEMESTER
Discipline Specific Elective Course (DSE) – MG5DSEZGY300, MG5DSEZIM300

BIOTECHNOLOGY: PRINCIPLES AND PRACTICES

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer **all** questions

Each question carries **1**marks

1. Select the enzyme which is known as "Molecular Glue".
   a) DNA Ligase b) RNA Polymerases
   c)DNA Polymerases d) Restriction Enzymes [U]/ [CO1]

2. BAC can accept insets of around ….....
   a) 100kb b) 10kb
   c) 30kb d) 300kb [U]/[CO1]

3. Name the technique separates DNA fragments primarily on the basis of size.

a) Northern blotting b) PCR

C)Southern blotting d) Agarose gel electrophoresis [U]/[CO1]

4. Primers in PCR determine the------

a) Target region to be amplified b) Colour of the DNA

c) Size of the gel d) Charge of the electrode [U]/[CO1]

5. Select the treatment commonly used to remove proteins during DNA extraction.

a) Proteinase treatment b) To separate DNA from cellular components

c) To sequence RNA d) To amplify DNA automatically [U]/[CO1]

6. The enzyme that joins the desired gene to the vector is-------

a) Restriction endonuclease b) Helicase

c) Primase d) DNA Ligase [U]/[CO1]

7. In colony hybridisation, bacterial colonies are transferred onto a ---------

a) Membrane such as nitrocellulose or nylon membrane b) Plastic plate

c) Glass slide d) Microtiter plate [U]/[CO1]

8. The chemical used in transfection is --------

a) DEAE Dextran b) EDTA

c) SDS d) Phenol [U]/[CO1]

9. Insulin is secreted by which cells of the pancreas?

a)a) Alpha cells b) Beta cells

c) Delta cells d) Acinar cells [K]/[CO3]

10. What is the primary goal of regenerative medicine?

a) Destroy damaged tissues b) Repair or replace damaged cells and tissues

c) Prevent cell division d) Increase drug toxicity [An]/[CO3]

[1x10=10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. Describe Cry proteins and their significance in Bt plants [K]/[CO3]

12. Discuss the role of growth hormone (somatotropin) in increasing milk production in cows [An]/[CO3]

13. Define bioremediation. Mention two advantages of bioremediation over conventional methods of

pollution control [An]/[CO3]

14. Define the terms ‘sticky ends’ and ‘blunt ends’ [K]/[CO1]

15. Define the term biotechnology and who coined the term Biotechnology? State the original context in which the term was used ? [K]/[CO1]

16. Write a short note on artificial chromosomes. [U]/[CO1]

17. Describe Northern blotting? Mention any two applications of Northern blotting. [U]/[CO1]

18. Explain DNA sequencing. Describe the principle of Sanger’s chain termination method and mention One application. [U]/[CO1]

19. What is proteome identification? Explain how proteins are identified and sequenced using proteomic techniques. [U]/[CO1]

20. Differentiate between gene gun and electroporation. [U]/[CO1]

21. Comment on media used in animal cell culture. [U]/[CO2]

22. What are stem cells? Mention their major uses. [U]/[CO2]

[3x10=30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Explain the production of transgenic mice and transgenic *C. elegance* and discuss their application in genetic engineering. [U]/[CO2]

24. What is CRISPR-Cas9? Explain its mechanism and application in genetic engineering [U]/[CO2]

25. Explain the role of biotechnology in human welfare, with special reference to the production and Application of of antibiotics and vaccines. [An]/[CO3]

26. Explain DNA finger printing and its application. [An]/ [CO3]

27. Explain the production of single cell protein through fermentation biotechnology and discuss its application. [An]/[CO3]

28. What are GMOs? Discuss their potential impacts on biodiversity. [U]/[CO4]

29. What is biopiracy? Explain its causes, effects and measures to prevent biopiracy with

suitable example. [U]/[CO4]

30. Explain the different types of IPR and their role in protecting biological resources and traditional knowledge. [U]/ [CO4]

[5x6=30]